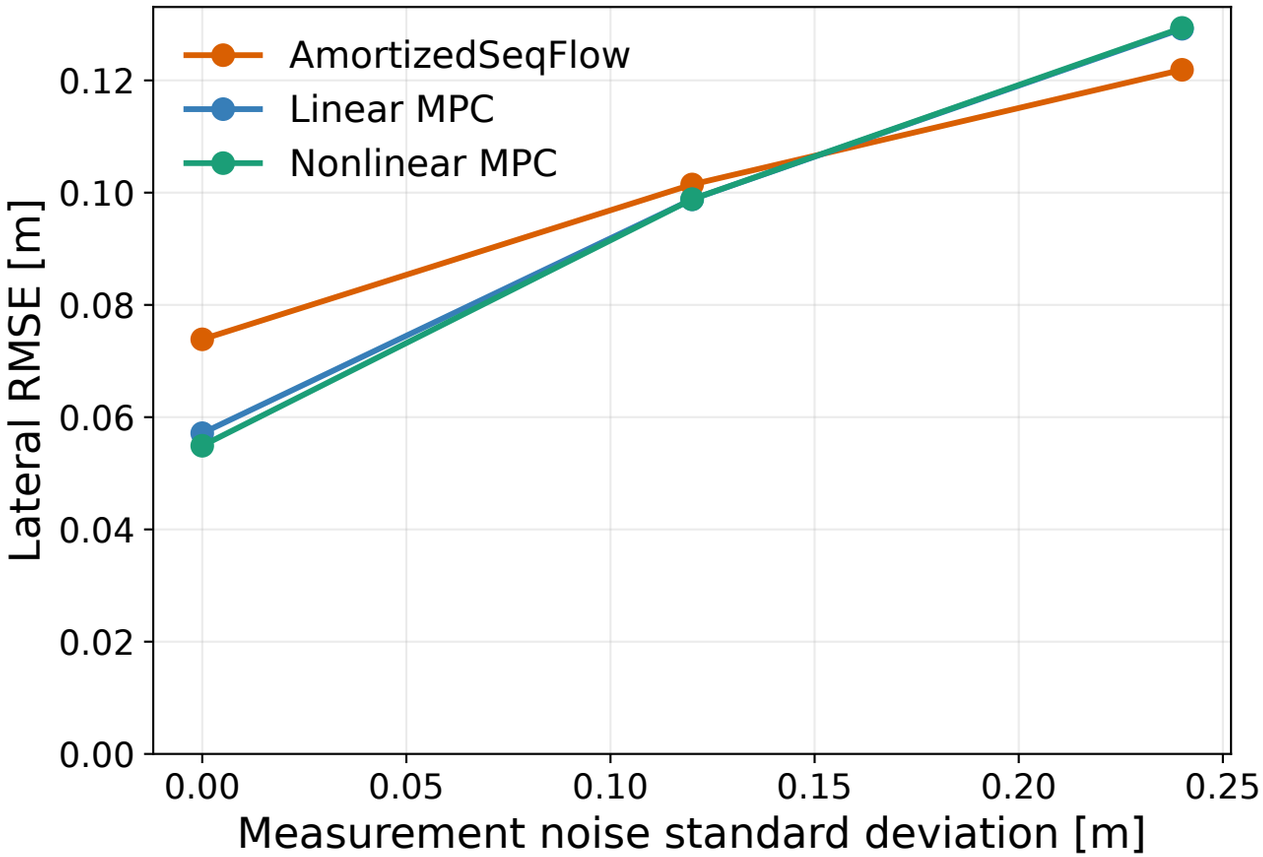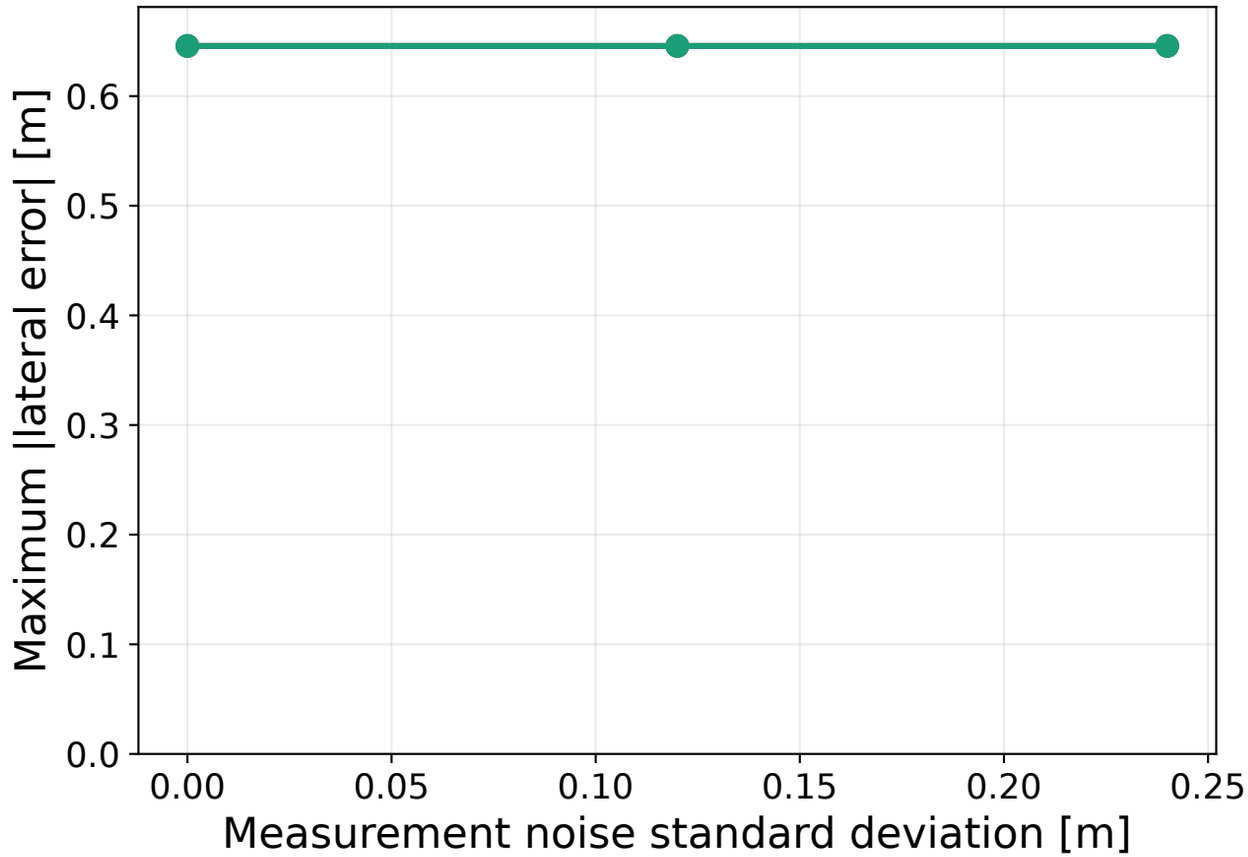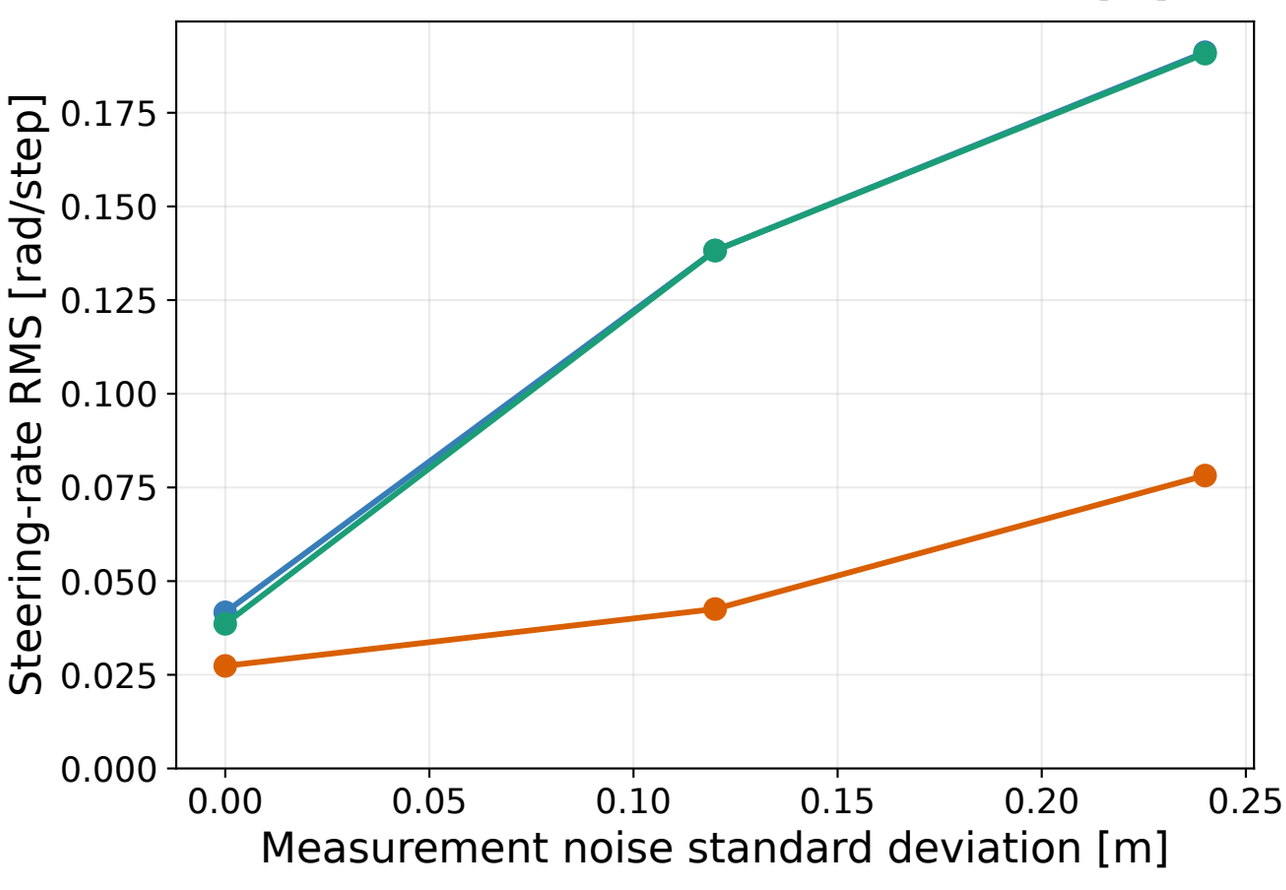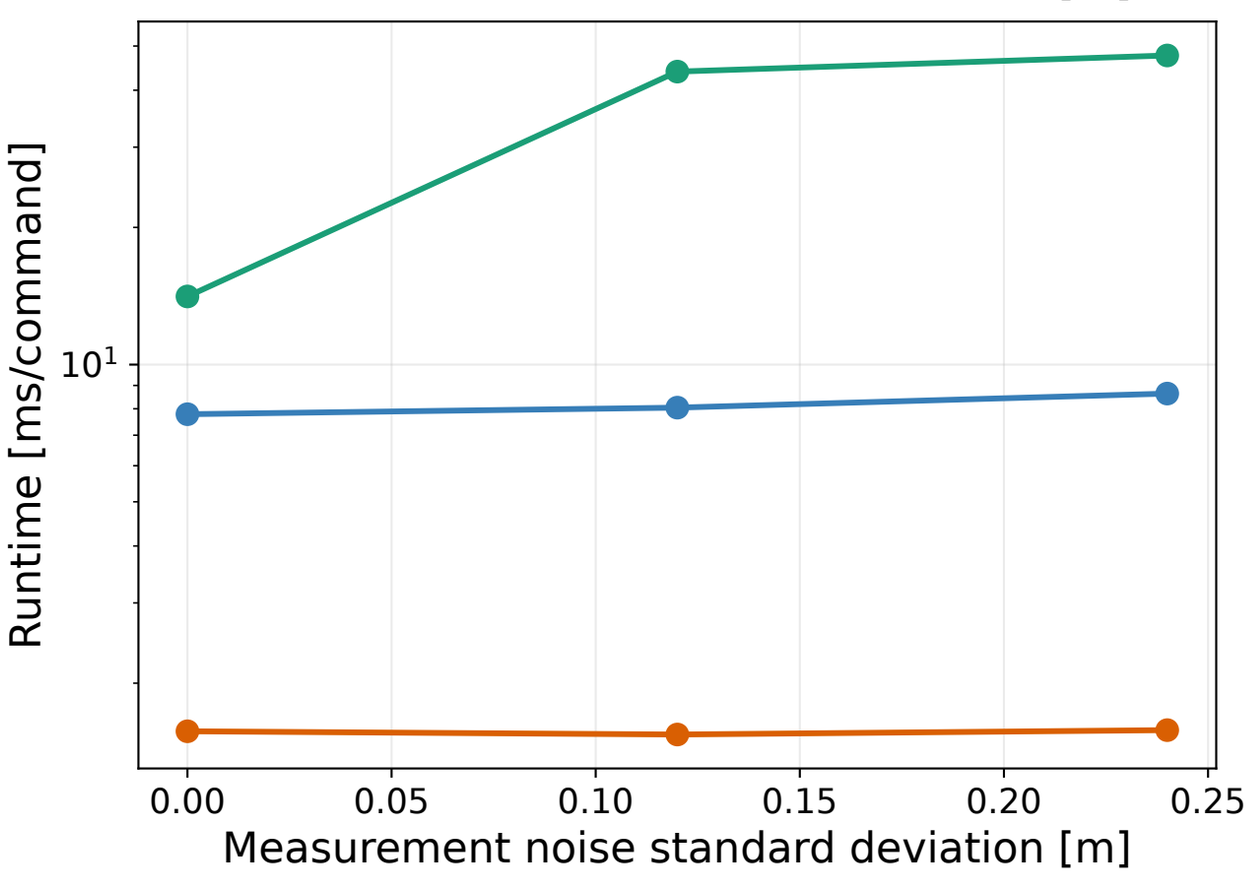Closed-loop performance under measurement noise